

Xianyuan Bao

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EDUCATION BACKGROUND

M.S. of Engineering, River Dynamics

Sep 2023 – Jun 2026

School of Civil and Hydraulic Engineering, Ningxia University

B.S. of Engineering, Building Environment and Energy Application Engineering

Sep 2018 – Jul 2022

School of Energy and Environment, Zhongyuan University of Technology

PROJECT EXPERIENCE

[Host] Coupling mechanisms of water-sediment-ecology-economy system in the Yellow River Basin based on multi-agent modeling

Postgraduate Innovation Project of Ningxia University (Grantee: Xianyuan Bao, 6,000 CNY) 2025 – 2026

This project aims to construct a coupling model of hydrological processes-vegetation succession-industrial activities, and deconstruct the complex mutual feedback mechanism of water and sediment migration, vegetation succession and industrial activities in the basin.

[Participate] Coordination mechanism and regulation strategy of multiple processes of water-sediment-ecology economy system in the Yellow River Basin

NSFC Programs for Joint Funds (Grantee: Prof. Enhui Jiang, 11,720,000 CNY) 2023 – 2026

The project explores the feedback mechanisms between environmental changes and socio-economic dynamics, with implications for long-term sustainable development in the region. As a key participant in this interdisciplinary project, I am mainly responsible for the research on the threshold for the development and utilization of water and sediment resources in different areas of the Yellow River Basin.

PUBLICATIONS

- [1] **Bao X.**, He, J., Wang, Z., Zhang, D., and Cui, X. 2025. “Research status, challenges and prospects of the utilization of Yellow River sediment resources”. *Journal of Sediment Research*, 50(04), 73-80.
- [2] He, J., **Bao X.**, Li, S., Zhang, D., Yang, H., Yang, Q., Li, X., Wen, X. 2026. “Deep learning-based response mechanism of nitrogen utilization, yield, and quality in mulched drip irrigation rice to nitrogen application rate”. *Agricultural Water Management*, 333(08), 110592.
- [3] **Bao X.**, He, J., Wang, Z., Zhang, D., Cui, X and Li, S. 2026. “Cross-Scale Coupling of Water-Sediment Resources and Socioeconomic System Interactions in the Yellow River Basin in China”. *Journal of Water Resources Planning and Management*, 152(09), 04026039.

PROFESSIONAL PRESENTATIONS

<i>[Oral Presentation]</i> Co-evolution mechanism of water-sediment resources and economic development from a governing perspective in the Yellow River Basin.	
<i>National Academic Forum for Hydraulic Engineering Postgraduates, Wuhan, China</i>	<i>May 29, 2025</i>
<i>[Oral Presentation]</i> A cross-scale coupling study on water-sediment and economic development in the yellow river basin based on urban agglomeration scale.	
<i>2nd Hydrological Process Forum, Guangzhou, China</i>	<i>Apr 20, 2025</i>
<i>[Oral Presentation]</i> Cross-scal ecoupling of water-sediment resources and socio-economic system interactions in the Yellow River Basin over different periods.	
<i>1st Innovation Academic Forum of Faculty of Engineering and Geography NXU, Yinchuan, China</i>	<i>Oct 17, 2024</i>

HONORS AND AWARDS

<i>Outstanding Master’s Degree Graduates</i>	<i>Jun 2026</i>
<i>2nd prize in the 11th China College Students’ Statistical Modeling Competition</i>	<i>Aug 2025</i>
<i>3rd prize in National Academic Forum for Hydraulic Engineering Postgraduates</i>	<i>May 2025</i>
<i>2nd prize in Innovation Academic Forum of Faculty of Engineering and Geography NXU</i>	<i>Nov 2024</i>
<i>Merit Student of Henan Province</i>	<i>Oct 2021</i>
<i>2nd prize in China Undergraduate Mathematical Contest in Modeling</i>	<i>Dec 2020</i>

GRANTS AND SCHOLARSHIPS

<i>Zhang Guangdou Outstanding Student Scholarship (8,000 CNY)</i>	<i>Apr 2026</i>
<i>Postgraduate Innovation Project of Ningxia University (6,000 CNY)</i>	<i>May 2025</i>
<i>Ningxia University Scholarships (once 2nd & once 3rd prize, 9,000 CNY)</i>	<i>Nov 2024, Dec 2025</i>
<i>Zhongyuan University of Technology Scholarships (once 2nd & twice 3rd prize, 2,000 CNY)</i>	<i>Dec 2018 – Dec 2021</i>

ACADEMIC SERVICES

Reviewer for Academic Journals:	
• <i>Water Resources Management</i>	<i>07/2025, 01/2026</i>
• <i>Geomorphology</i>	<i>05/2026</i>

OTHER EXPLORATIONS

<i>[Participate]</i> Field trip of watershed planning and management	
<i>Tsinghua University funded exploration</i>	<i>Jun 30, 2024 – Jul 06, 2024</i>